

Open to internship · March 2027

CONTACT

+33 6 51 75 16 99

nicolas.bailliet@etu.toulouse-inp.fr

Paris, France

github.com/baillietn

linkedin.com/in/nbailliet

hf.co/baillietn

baillietn.github.io

SKILLS

MACHINE LEARNING & LLMS

LLM architecture (GQA · RoPE · SwiGLU)	Proficient
Pre-training pipeline (data · curriculum · BPE)	Proficient
Fine-tuning (SFT)	Proficient
Mixed-precision training (BF16/FP8)	Intermediate
Inference & deployment (HF)	Intermediate

NUMERICAL & OPTIMIZATION

Numerical optimization (constrained)	Advanced
Optimal control · ODE solvers (shooting · MPC)	Proficient

FRAMEWORKS & TOOLS

`Python` `PyTorch` `HuggingFace` `Transformer Engine` `Flash Attention` `FP8 / BF16` `JAX/Flax` `NumPy` `scikit-learn` `Julia` `SQL` `Git` `Linux` `HPC`

LANGUAGES

French: Native

English: Professional

Nicolas Bailliet

LLM engineer

- Seeking a 6-month end-of-studies internship from mid-March 2027 · Based in Paris · Open to relocation · EU citizen
- LLM engineering & agentic systems, MLOps, model fine-tuning

PROFESSIONAL PROFILE

Final-year Master's student in Computer Science at ENSEEIHT, with a background in mathematics and numerical optimization. Built **HilbertLM** end to end **on a single desktop GPU**: custom tokenizer, pre-training on 20B tokens, fine-tuning and a live chatbot. Looking to build and deploy **LLM and agentic systems** in production, and to **fine-tune models** for the teams that use them.

PROFESSIONAL EXPERIENCE

Research Engineer Intern — Optimal Control & Scientific Computing

Jun–Aug 2025

IRIT / ENSEEIHT · Toulouse

`Julia · JuMP`

- Enhanced and validated an open-source Julia optimal-control solver library, across two axes: software maintenance (bug fixing) and numerical validation.
- Built an automated benchmark looping over 15 control problems: convergence checks for both the solver and JuMP (reference Julia modeling language), initialization comparison, and relative error in the L^2 norm on the state x and control u between the two solvers.
- Investigated an Earth-to-Moon transfer modeled as the Restricted Three-Body Problem (CR3BP); implemented regularization to smooth singularities and scoped solver limitations for future work.
- Contributed unit tests & API documentation via GitHub (commits, Pull Requests, peer reviews), positive team feedback for surfacing critical bugs.

EDUCATION

Diplôme d'Ingénieur (MSc equivalent), Computer Science & Engineering — ENSEEIHT (INP Toulouse)

2023–2027 (expected)

ML algorithms, numerical methods, optimization, HPC, big data systems. Includes a required 6-month end-of-studies internship.

Classes Préparatoires (CPGE), Mathematics & Physics — Lycée Chaptal, Paris

2020–2023

Intensive undergraduate program in mathematics and physics, preparing for the competitive entrance exams to French engineering schools.

PROJECTS

★ HilbertLM (135M & 163M)

Live demo ↗ baillietn.github.io/HilbertLM-Lab
HuggingFace ↗ huggingface.co/collections/baillietn/hilbertlm

`135M & 163M params` `20B tokens trained` `~70h · RTX 5080` `79K tok/s throughput` `4 models on HF Hub`

`Python · PyTorch · Transformer Engine · Flash Attention 3 · FP8 · HuggingFace · BPE`

Two decoder-only transformers built entirely from scratch. 30-layer architecture: GQA (9Q/3KV heads), RoPE, SwiGLU — custom BPE tokenizer (49K vocab). 4-stage curriculum pre-training (FineWeb-Edu, Python-Edu, FineMath, Cosmopedia-v2) + SFT on 100M conversational tokens (SmolTalk). Loss landscape visualization via PCA on checkpoints. Deployed as Base & Chat variants (BF16 + FP8) on HuggingFace Hub with streaming chatbot interface.

Soprasteria-ENSEEIHT · Water Quality IBD Prediction

`Python · TensorFlow/Keras · scikit-learn · pandas`

Neural network predicting Biological Diatom Index for river ecosystem monitoring. 70+ environmental features from Naïades DB (2000-2025); BatchNorm & Dropout; anomaly detection & feature importance analysis.

ENM-ENSEEIHT · Deep Learning Practical Modules

`Python · PyTorch · Jupyter`

Completed 8 progressive ML modules covering: NN fundamentals, CNNs, attention mechanisms, transformers, semantic segmentation, GANs/WGAN, quantile regression, learning-to-rank (RankNet + curriculum learning).

Stochastic Filtering & Data Assimilation

`Python · JAX · Flax · NumPy`

4 notebooks: Kalman Filter, Ensemble Kalman Filter (Lorenz dynamics), score-based diffusion models (JAX/Flax), PageRank via Markov chains.

Optimal Control — Course Materials & Orbital Transfer

`Julia · DifferentialEquations.jl · NLsolve.jl · ForwardDiff.jl`

6 practical modules: ODEs, shooting methods, MPC, automatic differentiation, Runge-Kutta schemes. Orbital transfer optimization (Hohmann transfer, 3D trajectory modeling).